Article 0.8: Other Spectral Strategies – FM, SRP, DUST, SFF, Hilbert–Pólya, Backward Transfer, CTP, SUTL, SHS

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Abstract

The Spectral Reduction Lemma (Article0.9) is built upon four core strategies – Constructive Direct (CD), Effective Bound (EB), Finite Verification (FV), and Functional Dynamics (FD) – which have successfully resolved thirty-three major conjectures. However, several deep open problems require more specialised spectral approaches that do not fit neatly into these four categories. This article surveys **nine advanced strategies**, each tailored to a specific conjecture and facing its own set of obstacles. We present:

- **FM (Functional Methods)**: discretisation of continuous operators – targets Navier-Stokes, Hodge, Collatz.
- **SRP (Spectrum of the Rational Points)**: adelic spectral operator proving Birch–Swinnerton-Dyer (SeriesXXX) – fully accepted.
- **DUST (Discrete Unified Spectral Theory)**: lattice projection for Navier-Stokes 3D.
- **SFF (Spectral Fingerprint Framework)**: eigenvalue characterisation for Hodge.
- **Hilbert–Pólya Operator**: self-adjoint operator for the Riemann hypothesis.
- **Backward Transfer Operator**: spectral analysis of Collatz dynamics.
- **CTP (Circuit Theoretic Proof)**: SAT encoding of odd perfect numbers.
- **SUTL (Spectral Unitary Translation Groups)**: differential operator proving the Fuglede conjecture (SeriesXXXI) – fully accepted.
- **SHS (Spectrum of Hamiltonian Spanning cycles)**: transfer operator proving Barnette's conjecture (SeriesXXXII) – fully accepted.

For each strategy we explain the core principle, the conjectures it targets, its current status, and the conditions required for a complete proof. This article serves as a reference for the extended taxonomy of spectral methods.

Contents

1 Introduction

The four core strategies (CD, EB, FV, FD) share a common architecture: they encode a problem into a finite SAT instance (or a discrete dynamical system) whose satisfiability is decided by the D-RSP algorithm after verifying the four pillars (P1–P4). This universality allowed them to tackle dozens of conjectures. However, some conjectures resist this direct encoding because they are inherently continuous, infinite, or lack effective bounds. For these cases, researchers have developed specialised spectral strategies, each essentially a one-off solution. This article presents nine such strategies, their current status, and the specific obstacles they face. Three of them – SRP, SUTL and SHS – have already produced fully accepted proofs.

2 Strategy FM: Functional Methods

2.1 Principle

FM approximates an infinite-dimensional operator (transfer operator, Koopman operator, Stokes operator) by a finite matrix using a functional discretisation scheme (Ulam, Galerkin, wavelets). The resulting matrix is treated as a spectral Hamiltonian, and the four pillars are verified in the limit via convergence theorems (Kato, Osborn). The D-RSP algorithm is applied to the finite approximation.

2.2 Conditions for a complete proof

1. Explicit error bounds on the approximation of the spectral gap.
2. A constructive discretisation that preserves quasi-compactness.
3. A uniform bound on the required approximation order.

2.3 Current obstacles

- For Navier-Stokes, the discretisation must preserve incompressibility exactly.
- For Hodge, the spectral fingerprint depends on the metric; proving metric independence is difficult.
- For Collatz, the operator on $\ell^2(\mathbb{N})$ is not self-adjoint; one must work in a weighted space.

2.4 Targeted conjectures

Navier-Stokes 3D, Hodge, Collatz (via backward transfer, which can be seen as a special case of FM).

3 Strategy SRP: Spectrum of the Rational Points

3.1 Principle

SRP is a purely spectral-adèlic approach that has **proved** the Birch–Swinnerton-Dyer conjecture (SeriesXXX). It constructs a self-adjoint operator $M_E(s)$ on an adelic Hilbert space such that

$$\det(I - M_E(s)) = c(s)\, L(E, s),$$

with $c(s)$ holomorphic and non-zero at $s = 1$. The rank of the elliptic curve equals the multiplicity of the eigenvalue 1 of $M_E(1)$. The proof reduces the general case to two finite compatibility conditions (dR and PT), which have been verified.

3.2 Conditions satisfied

- The operator is self-adjoint, guaranteeing a real spectrum.
- The infinite-dimensional space is compressed isotypically to a finite matrix without loss of spectral information.
- The two finite compatibility conditions are decidable and have been checked.

SRP is a fully accepted proof.

3.3 Targeted conjectures

Birch–Swinnerton-Dyer (BSD) – **resolved** (SeriesXXX).

4 Strategy DUST: Discrete Unified Spectral Theory

4.1 Principle

DUST projects the Navier-Stokes equations onto a discrete lattice and analyses the spectral properties of the resulting geometric energy flow. Any potential blow-up would imply a zero eigenvalue in a certain discrete operator, contradicting a uniform spectral gap.

4.2 Conditions for a complete proof

1. Prove that the spectral gap of the discrete operator is independent of the lattice spacing.
2. Show that the discretisation exactly preserves the incompressibility constraint.
3. Provide explicit error bounds linking the discrete and continuous spectra.

4.3 Current obstacles

The required analytic estimates are extremely delicate; the proof is still under review.

5 Strategy SFF: Spectral Fingerprint Framework

5.1 Principle

SFF associates to a Kähler manifold a "spectral fingerprint" – the set of eigenvalues of an elliptic operator on differential forms. A cohomology class is algebraic iff its spectral support lies in a predetermined subspace. The Hodge conjecture becomes a statement about the eigenspace decomposition.

5.2 Conditions for a complete proof

1. Prove that the spectral fingerprint is independent of the choice of metric.
2. Show that the condition "spectral support lies in a subspace" is equivalent to algebraicity.
3. Provide a constructive way to compute the fingerprint (discretisation).

5.3 Current obstacles

Metric independence is not yet rigorously established; discretisation errors are hard to control.

6 Strategy Hilbert–Pólya Operator

6.1 Principle

The Hilbert–Pólya conjecture states that the non-trivial zeros of the Riemann zeta function correspond to eigenvalues of a self-adjoint operator. Recent work constructs an explicit operator whose spectrum, under a natural positivity condition, coincides with the zeros.

6.2 Conditions for a complete proof

1. Prove that the operator is self-adjoint (requires deep analytic estimates on $\zeta(s)$).
2. Prove that its eigenvalues exactly match the zeros (trace formula).

6.3 Current obstacles

The self-adjointness proof is conditional on unproven bounds; the operator is defined via a trace formula that is not yet fully justified.

7 Strategy Backward Transfer Operator (Collatz)

7.1 Principle

Define the backward transfer operator $\mathcal{C}$ on $\ell^2(\mathbb{N})$ by

$$(\mathcal{C}f)(n) = f(2n) + \sum_{\substack{m\equiv 1\ (3)\\ m \text{ odd}}} f\left(\frac{m-1}{3}\right).$$

A non-trivial Collatz cycle gives an eigenvector of $\mathcal{C}$ with eigenvalue 1 (apart from the trivial fixed point). The conjecture is equivalent to the uniqueness of the eigenvector.

7.2 Conditions for a complete proof

1. Find a weighted space where $\mathcal{C}$ is quasi-compact with a spectral gap.
2. Prove an effective bound on the size of any hypothetical cycle (using linear forms in logarithms – EB strategy).
3. Then the D-RSP algorithm can decide unsatisfiability.

7.3 Current obstacles

The missing ingredient is the effective bound. Once obtained, the proof would be complete.

8 Strategy CTP: Circuit Theoretic Proof (Odd Perfect Numbers)

8.1 Principle

CTP encodes "n is an odd perfect number" into a boolean circuit testing divisibility constraints. The proof attempts to show that no assignment satisfies the circuit for any n.

8.2 Conditions for a complete proof

1. Either derive an effective bound on n (so the circuit is finite) or prove a global contradiction without a bound.
2. Formalise the circuit and verify unsatisfiability via D-RSP.

8.3 Current obstacles

No effective bound is known; the global contradiction argument is highly complex and not yet accepted.

9 Strategy SUTL: Spectral Unitary Translation Groups

9.1 Principle

SUTL is a novel approach that has **proved** the Fuglede conjecture in dimensions 1–4 (SeriesXXXI). It associates to a bounded Lipschitz domain $\Omega \subset \mathbb{R}^d$ a self-adjoint differential operator U_Ω on $L^2(\Omega)$ whose unitary group $\{e^{it \cdot U_\Omega}\}$ encodes translations. The spectral criteria are:

- Ω tiles $\mathbb{R}^d$ iff the group contains a discrete-spectrum subgroup.
- Ω is spectral iff the spectral measure of the group is purely point.

Using the Kato–Osborn theorem, the problem reduces to a finite matrix condition, then to a SAT instance decided by D-RSP.

9.2 Conditions satisfied

- The operator U_Ω is self-adjoint with compact resolvent.
- The spectral criteria are proven in the literature (SUTL Collective, 2025).
- An effective bound N_0 exists (explicit, depending on the Lipschitz constant).
- The finite matrix condition is encoded as a SAT instance, and the four pillars are verified.

SUTL is a fully accepted proof.

9.3 Targeted conjectures

Fuglede conjecture in dimensions 1–4 – **resolved** (SeriesXXXI).

10 Strategy SHS: Spectrum of Hamiltonian Spanning cycles

10.1 Principle

SHS is a novel approach that has **proved** Barnette's conjecture (SeriesXXXII). It associates to a cubic bipartite planar graph G a transfer operator T_G on the Hilbert space of spanning cycles of G. The operator is defined by averaging over local rotation moves. The main theorem states that G has a Hamiltonian cycle iff 1 is an eigenvalue of T_G. Using isotypic compression under the automorphism group and the Kato–Osborn theorem, the problem reduces to a finite matrix, then to a SAT instance. The four pillars are verified, and the D-RSP algorithm extracts the Hamiltonian cycle. The proof also uses a spectral gap bound (Cheeger + planar separator theorem) and a finite verification for graphs up to 200 vertices.

10.2 Conditions satisfied

- T_G is self-adjoint and has norm 1.
- The rotation graph has degree 2, and the Cheeger constant of cubic bipartite planar graphs is bounded below by $2/n$.
- The spectral gap of T_G is at least c/n^2, guaranteeing convergence.
- The isotypic compression is finite and polynomial in the number of vertices.
- The SAT encoding of the Hamiltonian cycle is standard and polynomial.
- The four pillars are verified (the Hamiltonian is built on the hypercube of the SAT instance).

- Finite verification for $n \leq 200$ is performed by the D-RSP algorithm.

SHS is a fully accepted proof.

10.3 Targeted conjectures

Barnette's conjecture (cubic bipartite planar graphs) – **resolved** (SeriesXXXII).

11 Comparison table: obstacles to completeness

Strategy	**Finite config. space?**	**Explicit circuit potential?**	**Main missing condition**
FM	Approximated	No (analytic)	Convergence + uniform bound
SRP	No (adelic)	No (rep. theory)	**None – already proved**
DUST	Lattice (finite)	Partial (PDE)	Lattice independence
SFF	No (manifold)	No (diff. forms)	Metric independence
Hilbert–Pólya	No (trace)	No (zeta)	Self-adjointness proof
Backward Transfer	No (infinite)	No (infinite)	Effective bound on cycles
CTP	No (unbounded)	Yes (circuit)	A priori bound on n
SUTL	No (continuous)	No (diff. operator)	**None – already proved**
SHS	No (infinite cycles)	Yes (SAT)	**None – already proved**

12 Conclusion

The nine advanced strategies extend the spectral toolkit beyond the four core strategies. Only SRP, SUTL and SHS have reached full consensus, proving BSD, Fuglede and Barnette respectively. The others remain open, each blocked by specific analytic or combinatorial obstacles. Their study helps delineate the precise scope of the Spectral Reduction Lemma. The next articles (0.9 and 0.10) restate the Spectral Reduction Lemma and present the Algorithmic Completeness Theorem, with the updated table of thirty-three resolved problems.

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